

Maps Open Ended Assignment

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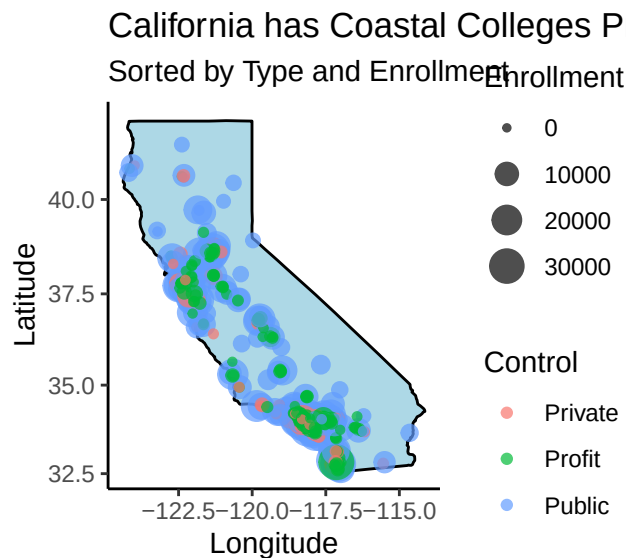
Graph 1: US Map (labels)

```
data(CollegeScores)
california_colleges <- CollegeScores %>% filter(State == "CA") %>% mutate(Name = str_to_title(Name))
california_map <- map_data("state", "california")
head(california_map)

##           long      lat group order   region subregion
## 1 -120.0060 42.00927     1     1 california      <NA>
## 2 -120.0060 41.20139     1     2 california      <NA>
## 3 -120.0060 39.70024     1     3 california      <NA>
## 4 -119.9946 39.44241     1     4 california      <NA>
## 5 -120.0060 39.31636     1     5 california      <NA>
## 6 -120.0060 39.16166     1     6 california      <NA>

#California Colleges
#gf_polygon(lat ~ long, data = california_map, group = ~group, fill = "lightblue", color = "black") %>%
ggplot() + geom_polygon(data = california_map, aes(x = long, y = lat, group = group), fill = "lightblue")

## Warning: Removed 53 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Graph 2: World Map (dots)

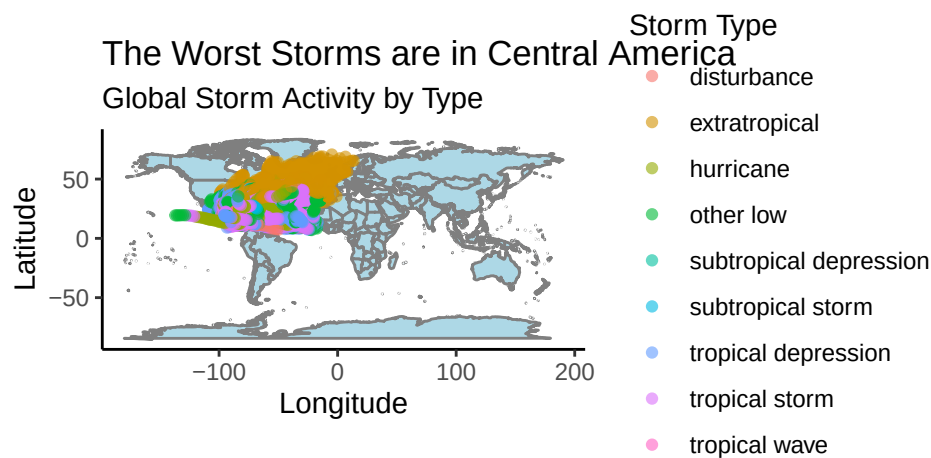
```
world <- map_data("world")
data(storms)
head(storms)
```

```
## # A tibble: 6 x 13
##   name   year month   day hour   lat   long status      category  wind pressure
##   <chr> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <fct>      <dbl> <int>    <int>
## 1 Amy    1975     6    27     0  27.5 -79 tropical de~      NA     25     1013
## 2 Amy    1975     6    27     6  28.5 -79 tropical de~      NA     25     1013
## 3 Amy    1975     6    27    12  29.5 -79 tropical de~      NA     25     1013
## 4 Amy    1975     6    27    18  30.5 -79 tropical de~      NA     25     1013
## 5 Amy    1975     6    28     0  31.5 -78.8 tropical de~      NA     25     1012
## 6 Amy    1975     6    28     6  32.4 -78.7 tropical de~      NA     25     1012
## # i 2 more variables: tropicalstorm_force_diameter <int>,
## #   hurricane_force_diameter <int>
```

```
#Global Storms
```

```
#world %>% gf_polygon(lat ~ long, group = ~group, fill = "lightblue", color = "gray50") %>% gf_point(lat
```

```
ggplot() +
  geom_polygon(data = world, aes(x = long, y = lat, group = group),
    fill = "lightblue", color = "gray50") +
  geom_point(data = storms, aes(x = long, y = lat, color = status), alpha = 0.6) +
  coord_quickmap() +
  labs(title = "The Worst Storms are in Central America",
    subtitle = "Global Storm Activity by Type",
    x = "Longitude", y = "Latitude",
    color = "Storm Type") #ChatGPT fixed the never ending error I was getting in aesthetics
```



I'm going to go on a whim and assume something is wrong with the dataset as there is no storms besides in the Atlantic Ocean and neighboring areas. This is just false as there is currently typhoons/storms throughout Asia.

Graph 3: Any Map (Filled)

```
data(world.cities)
head(world.cities)
```

```
##           name country.etc  pop  lat  long capital
## 1 'Abasan al-Jadidah  Palestine  5629 31.31 34.34      0
## 2 'Abasan al-Kabirah  Palestine 18999 31.32 34.35      0
## 3 'Abdul Hakim      Pakistan 47788 30.55 72.11      0
## 4 'Abdullah-as-Salam  Kuwait  21817 29.36 47.98      0
## 5 'Abud              Palestine  2456 32.03 35.07      0
## 6 'Abwein           Palestine  3434 32.03 35.20      0
```

```
country_pop <- world.cities %>% group_by(country.etc) %>% summarise(pop = sum(pop)) %>% mutate(region =
world_pop_map <- right_join(country_pop, world, by = "region")
#World Pop by Country
gf_polygon(lat ~ long, data = world_pop_map, group = ~group, fill = ~pop, color = "gray50") %>% gf_refi
```

Asia is the Most Densely Populated

World Population by Country

